

AUTOMATIC MEDICINE REMAINDER USING LPC2148

REPORT

Course Name – ARM

Abstract

Medication adherence is a significant challenge, particularly for elderly individuals and patients who must follow strict medication schedules. Missing prescribed doses can reduce treatment effectiveness and lead to health complications. To address this problem, an Automatic Medicine Reminder System was developed using the LPC2148 ARM7 microcontroller.

The system utilizes the Real-Time Clock (RTC) module of the LPC2148 to maintain accurate time and compare it with user-configured medicine schedules. Alarm timings are entered through a keypad and stored in memory. When the current time matches a scheduled alarm, the system generates alerts using a buzzer and visual indicators while displaying relevant information on a 16×2 LCD. Additional features such as alarm viewing, addition, deletion, snooze functionality, and alarm acknowledgement are incorporated to improve user interaction and flexibility.

The developed system successfully provides reliable medication reminders at scheduled intervals and offers a simple user interface for alarm management. The implementation demonstrates the effective integration of RTC, LCD, keypad, ADC, and interrupt-based control mechanisms within an embedded system, resulting in a practical and cost-effective solution for assisting users in maintaining regular medication schedules.

Introduction

In modern healthcare, timely and consistent medication intake plays a vital role in ensuring effective treatment and patient well-being. However, many individuals, particularly elderly patients, people with chronic illnesses, and those following multiple medication schedules, often forget to take their medicines at the prescribed times. Such missed doses can reduce treatment effectiveness and may lead to serious health

complications. Therefore, there is a growing need for an automated system that can assist users in managing their medication schedules accurately and reliably.

The Automatic Medicine Reminder System is designed to address this problem by providing timely reminders based on user-defined schedules. The system utilizes the LPC2148 ARM7 microcontroller along with its Real-Time Clock (RTC) module to maintain accurate time and monitor scheduled medication timings. A keypad is used to configure alarms, while a 16×2 LCD provides user interaction and status information. When a scheduled time is reached, the system generates audio and visual alerts to notify the user.

The primary objectives of this project are to develop a reliable medicine reminder system, provide a simple user interface for alarm management, enable the addition, viewing, and deletion of medicine schedules, and ensure accurate real-time monitoring using the RTC module. The system aims to offer a cost-effective and user-friendly solution that helps users maintain medication adherence and improve healthcare management.

Problem Statement

Many individuals, especially elderly patients and people undergoing long-term medical treatment, are required to take medicines at specific times throughout the day. Due to busy schedules, memory-related issues, or lack of an effective reminder mechanism, patients may forget to take their medications on time. Missing or delaying prescribed doses can reduce treatment effectiveness, worsen health conditions, and increase the risk of complications.

Traditional reminder methods such as written schedules, manual alarms, or mobile phone notifications may not always be reliable or convenient for all users. Therefore, there is a need for a dedicated, user-friendly, and reliable system that can automatically monitor time, remind users when medication is due, and provide simple alarm management features.

The problem addressed in this project is the development of an embedded system capable of maintaining real-time schedules, generating timely alerts for medication intake, allowing users to configure and manage multiple alarms, and providing clear visual and audible notifications to ensure that medicines are taken at the correct time.

System Overview

The Automatic Medicine Reminder System is an embedded application built around the LPC2148 ARM7 microcontroller. The system operates by accepting user-defined medicine schedules, continuously monitoring the current time using the Real-Time Clock (RTC), and generating alerts whenever a scheduled medication time is reached.

Input Stage:

The user interacts with the system through a keypad to set the current time, add medicine alarms, view stored alarms, and delete existing alarms. Temperature data is obtained from a temperature sensor connected to the ADC channel of the LPC2148.

Processing Stage:

The LPC2148 microcontroller acts as the central processing unit of the system. It stores alarm information, maintains real-time tracking through the RTC module, compares the current time with stored alarm schedules, and processes user commands received from the keypad. The controller also monitors alarm status and manages snooze and alarm acknowledgement functions.

Output Stage:

The system provides information to the user through a 16×2 LCD and UART terminal. When a scheduled medicine time is reached, the controller activates visual and audible alerts using LEDs and a buzzer. The LCD displays alarm information, current time, and system status, while the UART interface provides detailed user interaction and monitoring.

Working Principle:

After initialization, the user sets the current time and configures medicine alarms. The RTC continuously updates the current time, and the microcontroller periodically compares it with the stored alarm schedules. When a match occurs, the system generates alerts and waits for user action such as alarm acknowledgement or snooze. This process continues continuously, ensuring timely medication reminders and effective schedule management.

Hardware Description

LPC2148 Microcontroller

The LPC2148 is a 32-bit ARM7TDMI-S based microcontroller developed by NXP Semiconductors. It serves as the central processing unit of the Automatic Medicine Reminder System. The controller features on-chip peripherals such as Real-Time Clock (RTC), Analog-to-Digital Converter (ADC), UART, I²C interface, timers, and external interrupts, making it suitable for embedded monitoring and control applications. In this project, the LPC2148 manages alarm scheduling, user interaction, time tracking, temperature monitoring, and alert generation.

Real-Time Clock (RTC)

The built-in RTC module of the LPC2148 is used to maintain accurate date and time information. It continuously tracks the current time and enables the system to compare stored alarm schedules with real-time values. The RTC forms the core of the reminder functionality.

16×2 LCD Display

A 16×2 LCD is used to display system information such as current time, temperature readings, alarm status, and user prompts. The display provides a simple and user-friendly interface for monitoring system operation.

Matrix Keypad

A 4×4 matrix keypad is used as the primary input device. It allows the user to enter the current time, add medicine alarms, delete alarms, and navigate through various system functions. The keypad provides a convenient method for user interaction without requiring a computer connection.

Temperature Sensor

A temperature sensor is interfaced with the ADC channel of the LPC2148 to measure ambient temperature. The analog output of the sensor is converted into digital values using the microcontroller's built-in ADC. The measured temperature is displayed on the LCD and can be viewed through the system menu.

Buzzer

A buzzer is used to generate audible alerts whenever a scheduled medicine alarm is triggered. The buzzer ensures that users receive clear reminders even when they are not actively monitoring the display.

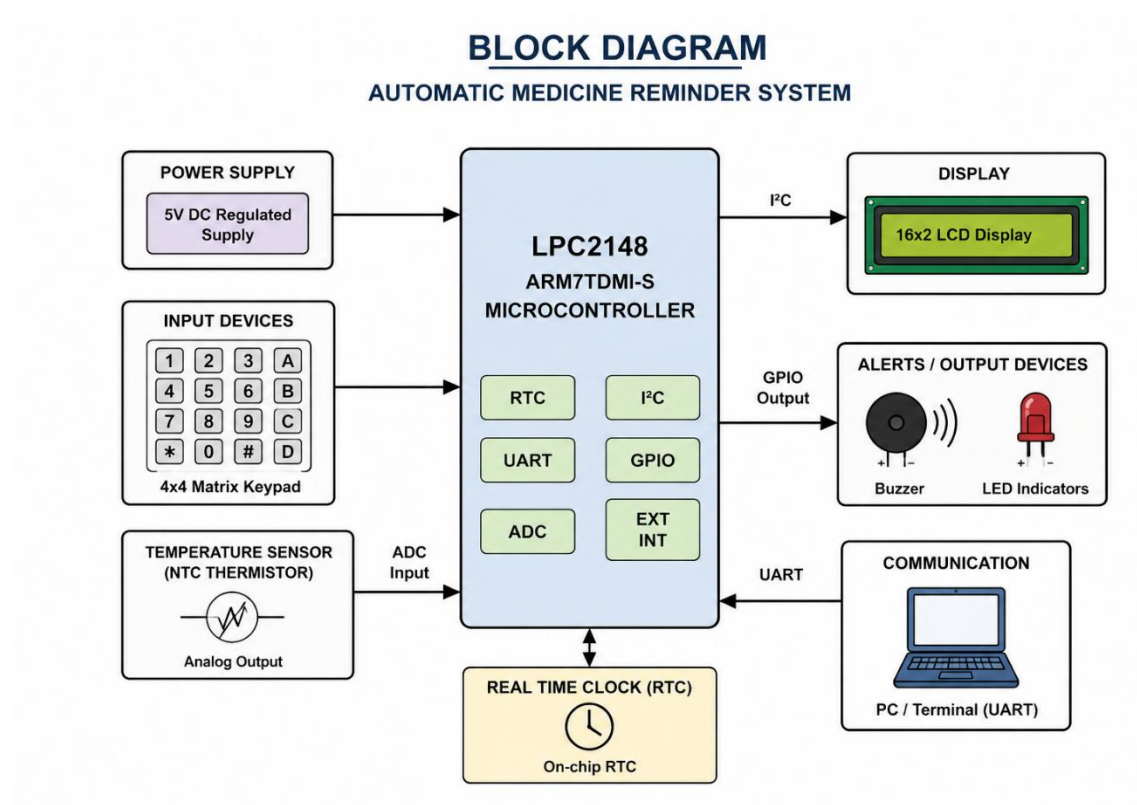
LEDs

LED indicators are used to provide visual notification of alarm events. When an alarm is triggered, the LEDs illuminate along with the buzzer, improving alert visibility and ensuring that reminders are noticeable in different environments.

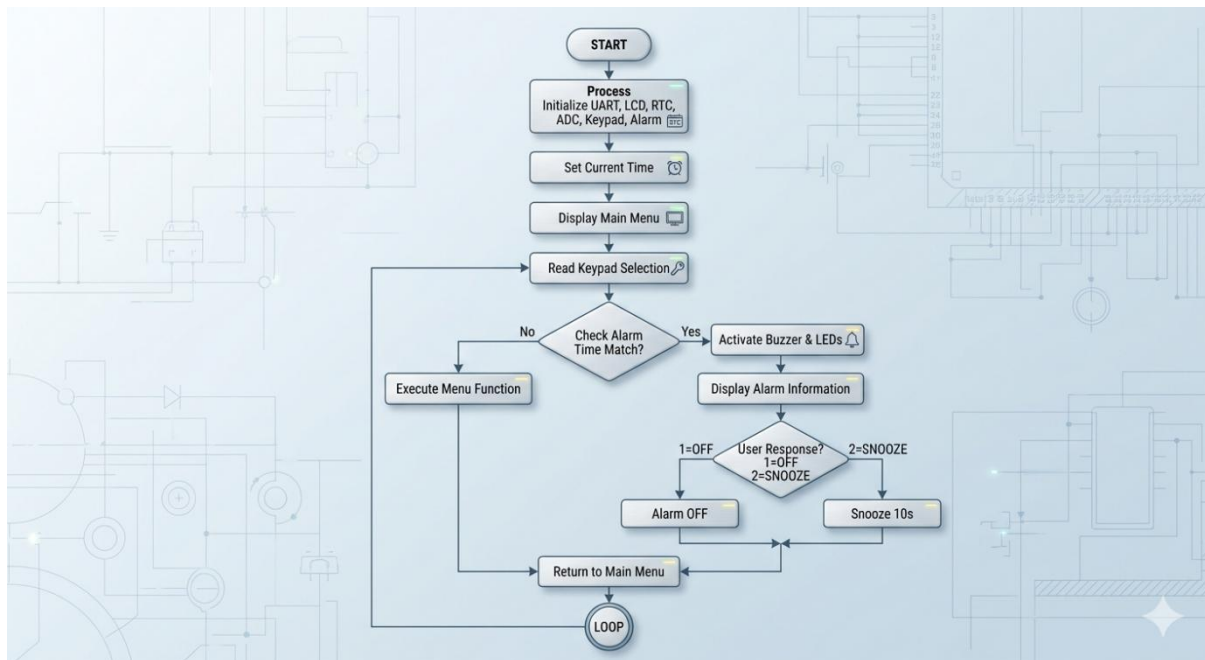
UART Interface

The UART communication interface is used for serial monitoring and debugging. It displays detailed system messages, alarm information, menu options, and user interactions through a terminal application connected to a computer.

Block diagram



Algorithm / Flowchart



Interfacing Details

The Automatic Medicine Reminder System is built around the LPC2148 microcontroller, and all peripherals are interfaced through its GPIO, UART, I2C, ADC, and RTC related pins. Proper pin mapping is essential to ensure reliable communication between the microcontroller and each external module. A common ground is maintained between the LPC2148 board and all connected devices.

The **UART0 interface** is used for serial communication with the PC terminal. In this project, UART0 is connected through **P0.0 (TXD0)** and **P0.1 (RXD0)**. This interface is used to display menu options, system messages, alarm information, and debugging output.

The **LCD display** is interfaced using the I2C bus. The LCD module is connected through **P0.2 (SDA)** and **P0.3 (SCL)**. Since the LCD is operated through an I2C backpack, the internal LCD control signals such as RS, EN, and data lines are managed through the I2C expander and do not require separate microcontroller pins.

The **4×4 matrix keypad** is used as the main input device for menu selection and time entry. The keypad rows are connected to **P1.20 to P1.23** as outputs, and the columns are connected to **P1.16 to P1.19** as inputs. Key scanning is performed by activating one row at a time and reading the corresponding column value.

The **7-segment display** is used to show the pressed key or entered digit during time setup. The segment data lines are connected to **P0.16 to P0.23**, while the

enable/control lines are connected to **P0.28 to P0.31**. These pins are configured as output lines and are driven according to the required digit pattern.

The **temperature sensor** is interfaced through the ADC channel of the LPC2148. The thermistor output is converted into a voltage using a resistor divider network, and this analog voltage is read by the microcontroller through the ADC input. The digitized value is then converted into temperature and displayed on the LCD and UART.

The **buzzer** is connected to a GPIO output pin and is activated whenever a medicine alarm is triggered. In addition, **LED indicators** are connected to GPIO pins and are used as visual alarm signals. The LEDs are configured in common-anode mode, so they glow when the output pin is driven LOW.

The **RTC module** is internal to the LPC2148 and does not require external pin connection. It maintains real-time tracking of the current time and is used to compare against the stored alarm schedule.

Overall, the interfacing arrangement integrates input devices, display units, alert mechanisms, and sensing modules into a single embedded system. This enables the medicine reminder system to operate as a compact and interactive real-time application.

Calculations & Specifications

RTC Configuration Calculation

The LPC2148 RTC requires the Prescale Integer Register (PREINT) and Prescale Fraction Register (PREFRAC) to generate an accurate 1-second time base from the RTC clock source.

For a 32.768 kHz RTC clock:

$\text{PREINT} = 456 = 0x01C8$

$\text{PREFRAC} = 25088 = 0x61C0$

These values are loaded into the RTC registers to ensure that the seconds, minutes, and hours counters increment accurately.

RTC Configuration Used:

$\text{PREINT} = 0x01C8$

$\text{PREFRAC} = 0x61C0$

$\text{CCR} = 0x01$

UART Baud Rate Calculation

UART communication is configured for 9600 bps.

Baud Rate Formula:

$$\text{Baud Rate} = \text{PCLK} / (16 \times \text{DLL})$$

Where:

$$\text{PCLK} = 15 \text{ MHz}$$

$$\text{DLL} = 98 (0x62)$$

$$\text{Baud Rate} = 15000000 / (16 \times 98)$$

$$\text{Baud Rate} \approx 9566 \text{ bps}$$

This value is sufficiently close to the standard baud rate of 9600 bps and provides reliable serial communication.

ADC Resolution Calculation

The LPC2148 contains a 10-bit ADC.

$$\text{Resolution} = 2^{10} = 1024 \text{ Levels}$$

ADC Output Range:

0 to 1023

Reference Voltage:

$$\text{VREF} = 3.3 \text{ V}$$

Voltage Calculation:

$$\text{Voltage} = (\text{ADC Value} \times 3.3) / 1023$$

Thermistor Voltage Calculation

The NTC thermistor is connected in a voltage divider configuration with a 10 kΩ resistor.

Sensor Voltage:

$$\text{Voltage} = (\text{ADC Value} \times 3.3) / 1023$$

Thermistor Resistance:

$$R_T = ((3.3 \times 10000) / \text{Voltage}) - 10000$$

Where:

RT = Thermistor resistance

10000 Ω = Fixed resistor value

Temperature Calculation

The Steinhart-Hart equation is used to convert thermistor resistance into temperature.

Temperature (Kelvin):

$$T = 1 / [A + B \ln(R) + C(\ln(R))^3]$$

Temperature (°C):

$$\text{Temperature} = T - 273.15$$

Constants used:

$$A = 0.001129148$$

$$B = 0.000234125$$

$$C = 0.0000000876741$$

The final result is displayed in degrees Celsius.

Alarm Specifications

Parameter	Specification
Number of Alarms	Up to 9
Time Format	24-Hour Format
Alarm Resolution	1 Minute
Snooze Duration	10 Seconds
Alarm Indication	Buzzer and LEDs
User Input	4×4 Matrix Keypad
Display	16×2 LCD and UART

System Specifications

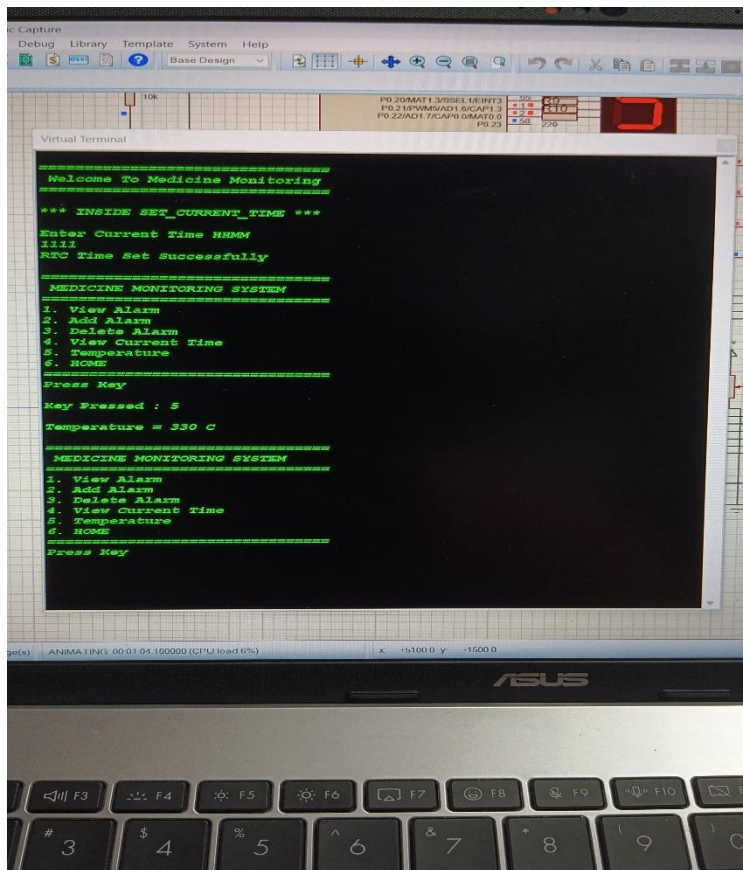
Parameter	Value
Microcontroller	LPC2148 ARM7TDMI-S
Supply Voltage	5 V DC
RTC Clock	32.768 kHz
ADC Resolution	10-bit
UART Baud Rate	9600 bps
LCD Interface	I ² C
Temperature Sensor	NTC Thermistor
Keypad Type	4×4 Matrix Keypad
Alert Devices	Buzzer and LEDs

These calculations and specifications determine the timing accuracy, temperature measurement capability, alarm operation, and overall performance of the Automatic Medicine Reminder System.

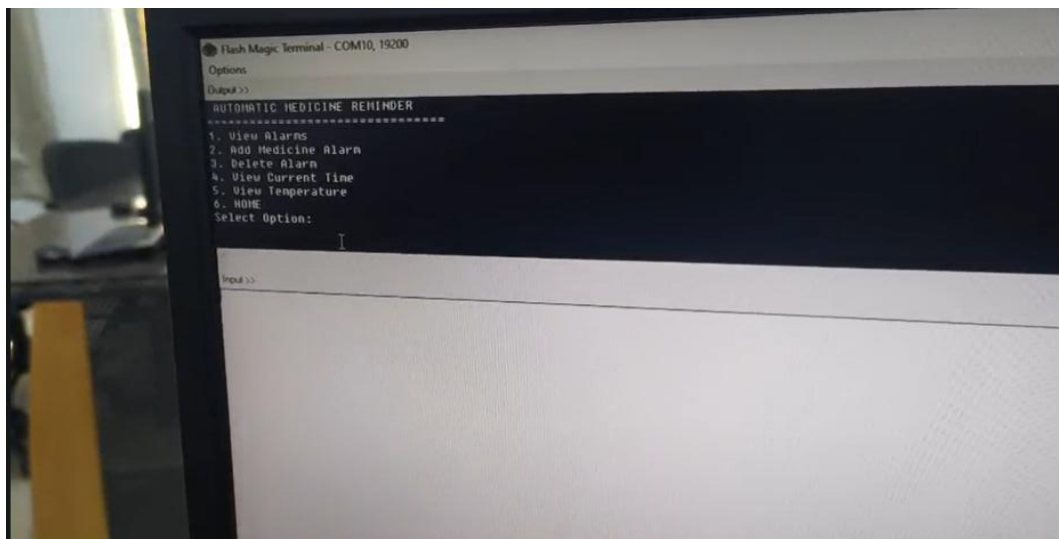
Implementation

The hardware implementation of the Automatic Medicine Reminder System is based on the LPC2148 ARM7 development board. A 16×2 LCD is interfaced through the I²C module for displaying system information, while a 4×4 matrix keypad is used for user input and menu navigation. A thermistor connected to the ADC channel enables temperature measurement. A buzzer and LEDs are used to generate audible and visual alerts whenever a scheduled medicine alarm is triggered. The built-in RTC of the LPC2148 maintains accurate time for alarm scheduling and monitoring.

The software implementation was developed using the Embedded C programming language. The program includes modules for RTC management, keypad scanning, LCD interfacing, UART communication, ADC-based temperature measurement, alarm scheduling, snooze functionality, and user menu handling. The code was written, compiled, and debugged using the Keil µVision IDE. The complete system was tested on the LPC2148 development board to verify alarm generation, user interaction, temperature monitoring, and real-time operation.



Results



Analysis

The system performed reliably in monitoring time and generating medicine reminders at the programmed intervals. The integration of RTC, LCD, keypad, ADC, and alarm mechanisms demonstrated the effectiveness of the LPC2148 microcontroller in real-time embedded applications. The implemented solution provides a simple, low-cost, and user-friendly approach for assisting users in maintaining regular medication schedules.

Conclusion

The Automatic Medicine Reminder System was successfully designed and implemented using the LPC2148 ARM7 microcontroller. The system effectively utilizes the Real-Time Clock (RTC) to monitor time and generate reminders at scheduled intervals. Through the integration of a keypad, LCD, buzzer, LEDs, UART communication, and temperature sensing, the system provides a complete and user-friendly solution for medication schedule management.

Advantages:

- Accurate real-time medication reminders using the RTC module.
- Simple user interface for adding, viewing, and deleting alarms.
- Audible and visual alerts improve notification reliability.
- Temperature monitoring capability through the ADC interface.
- Low-cost and compact embedded system implementation.

Limitations:

- Alarm data is not retained after power loss due to the absence of backup power or non-volatile storage.
- The system supports a limited number of alarms based on available memory.
- Medicine information such as dosage and drug name is not stored.
- Remote monitoring and mobile connectivity features are not included.

Overall, the project demonstrates the practical application of embedded systems in healthcare assistance and provides a reliable foundation for future enhancements such as battery backup, wireless communication, cloud integration, and advanced patient monitoring features.

Future Scope

The Automatic Medicine Reminder System can be further enhanced by incorporating a battery backup to retain time and alarm settings during power failures. Wireless communication technologies such as Bluetooth, Wi-Fi, or GSM can be added to enable remote monitoring and notification through mobile applications. Additional features

such as medicine name storage, dosage tracking, and cloud-based data logging can further improve the usability and effectiveness of the system.

References

1. LPC2148 User Manual, NXP Semiconductors.
2. LPC2148 Datasheet, NXP Semiconductors.
3. Keil μ Vision IDE Documentation.

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